

PSYC 210 – Midterm Study Guide

Chapter 1: Visualizing Data

Descriptive Statistics...describe a sample (observations!).

Example...Many UNC students are from North Carolina. 🚗

Inferential Statistics...make an inference (judgment!) about a sample.

Example...UNC students from North Carolina have significantly different food preferences than UNC students from other parts of the world. 🍏

Variable...something that can differ.

Example...Gender, age, hometown 😊

Measurement...orderly assign numbers to variables.

Example...You have a variable “happiness.” You *measure* it on a 1-5 Lickert scale. ✔

Data...observations of variables coded into categories.

Example...My variable is “happiness.” I observe people’s happiness by asking them to rate it on a 1-5 Likert scale. The responses I receive are data.

1	2
3	4

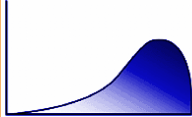
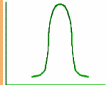
Discrete...finite number of values

Example...kids (you can’t have half a kid!) 📎

Continuous...infinite number of values

Example...height (x meters, x decimeters, x centimeters, etc.) 🧑

Some Ways Frequency Distributions Differ!

Central Tendency	Variability	Skewness	Kurtosis
“Central score”	“difference in scores”	“outliers”	“shape”
<u>Examples...</u>	<u>Examples...</u>	<u>Examples...</u> **Confusing!**	<u>Examples...</u>
1. Mean (minimizes deviations from the center) **most common**	1. Range (subtract lowest value from highest value)	1. Positive Skew 	1. Mesokurtic Curve 
2. Median (center score when order values lowest to highest)		2. Negative Skew 	2. Leptokurtic Curve 
3. Mode (most popular score)			3. Platykurtic Curve 

Types of Measurement!

Nominal “name”	Ordinal “Order”	Interval “Spacing”	Ratio “True Zero”
Numbers <i>don't</i> have meaning!	Numbers have meaning!	Numbers have meaning <u>AND</u> equal spacing between categories!	Numbers have meaning <u>AND</u> equal spacing <u>AND</u> true zero. (True zero is when none of the variables exists).
Example...Fruit. (Apples aren't “higher” than oranges on a number scale, they're just different!)	Example...World Population Rankings. (Numbers have meaning but spacing between each country's population is different!)	Example...Fahrenheit scale (Equal spacing b/t 1 degree and 2 degrees)	Example...Heart Rate (you can have zero).
			
<u>What graph to use?</u> 1. Bar Graph	<u>What graph to use?</u> 1. Bar graph	<u>What graph to use?</u> 1. Bar Graph	<u>What graph to use?</u> 1. Bar Graph
			
2. Pie Chart	2. Line graph	2. Histogram (bars touch)	2. Histogram (bars touch)
			
		3. Line graph	3. Line graph
			

Types of Frequency

Response	Frequency (number of responses)	Cumulative Frequency (number of responses <u>AND</u> add the previous scores)	Relative Frequency (%) (percent of responses)	Cumulative Relative Frequency (percent of responses <u>AND</u> add the previous scores)
Totally Against (0)	7	7	10	10
No Opinion (1)	19	26 (new score + 7)	27	37 (new score + 10)
Yes, but...(2)	25	51 (new score+26+7)	35	72 (new score + 37+10)
Totally Fair (3)	20	71 (new score+51+26+7)	28	100 (new score+72+37+10)

Class Interval...creating groups for the responses you receive based on defined ranges

Example...instead of Dr. Harrison reporting individual percentage grades, he use the class interval of 'A,' 'B,' 'C,' 'D,' and 'F' 

Lesson 2: Characteristics of Distributions

Mean...single number that minimizes deviations from itself

Equation...
$$\bar{X} = \frac{\sum_{i=1}^N X_i}{N}$$

Definition...average

Note...the sum of deviations from the mean is zero

Why Use the Mean?

- 1 good single descriptor
- 2 unbiased, good for comparing groups
- 3 use for *interval* or *ratio* scale

Median...middle point of data (splits data in half)

Why Use the Median?

- 1 when you have outliers (mathematically the mean is easily affected by outliers, but the median is much less affected!)
- 2 when you have ordinal data

Mode...most common response

Why Use the Mode?

- 1 when you have nominal data (i.e. data when the numbers don't have meaning!)
- 2 mode is the *least* affected by outliers

Variability...amount that scores differ from each other

Three Measures of Variability!

<u>Range</u> *pair with median*	<u>Variance</u> *don't worry about this yet*	<u>Standard Deviation</u> *pair with mean*
'Max - min = range'	'Average deviations around the mean'	'The square-root of variance' (measures data's skew from the mean!)
Simple subtraction!	$s^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$	$s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}$

Lesson 3: Relative Standing, Sampling Distributions, & Hypothesis Testing

Relative Standing...comparing groups of data to each other

Example...comparing how many likes you got on instagram in comparison to others 🤖

Percentile & Percentile Ranks...the percent of data that falls below your data point

Example...if you score in the 80th percentile on the SAT, then you beat 80% of testers 😊

Notation... $P_{.60}$ means that your data is the 60th percentile, so it beat 60% of the other data

Z score...a way to compare different distributions by creating a new, standardized Z distribution

Example...it's hard to compare if a golf player is “better” than a basketball player, but Z scores can tell you who is better 🤖

Method...Z scores make the means of the original distributions 0 and makes the standard deviations 1

Equation...

$$Z = \frac{X_i - \bar{X}}{s_x}$$

🤖 Note = the median of a Z score distribution is the 50th percentile AND area under the curve is 1.0

Steps to Using Z Scores...

- Step 1. Plug a data point into the Z score equation.
Step 2. You now have a Z score and its unit of measurement is in *standard deviations*
Step 3. Check the *giant* Z score table to find what percentile rank your Z score is associated with.
Done! 😊

Population distribution...a distribution of scores for *every* person in a group 🤖🤖

Example...if you got the GPA of every college student in the *entire* United States

Sample distribution...a distribution of scores for *some* people in a group 🤖

Example...if you got only the GPAs of UNC college students (not *every* college)

Sampling distribution...a distribution of means from many sample distributions 📊

Example...if you got the mean GPA of UNC students, the mean GPA of Duke students, and the mean GPA of NC state students. THEN, you make a graph where the data in the graphs are the means from these colleges.

Central Limit Theorem...the larger sample size you use on your **sampling** distribution, the more “normal” your curve becomes (your graph becomes more bell-curved shaped) ↻

Example...if you took the mean GPAs from 100 schools (instead of merely 5 schools) you have more data so the data will gravitate more toward an average looking curve

Standard Error of the Mean...the overall mean of a sampling distribution 🤖

Example...if you got the mean GPA of UNC students, the mean GPA of Duke students, and the mean GPA of NC state students. THEN, you make a graph where the data in the graphs are the means from these colleges. THEN, you get the mean of this new graph.

**Note: this shows the reliability of the mean of your sampling (*smaller* standard error of the mean scores indicate HIGHER reliability)

Hypothesis testing...tests if a certain mean score is or is not *significantly* different than population mean



Example...if you find the mean student GPA in our PSYC 210 class and compare it to the overall mean student GPA in every PSYC 210 class Dr. Harrison has ever taught AND THEN you test if our class's mean GPA is *significantly* different than the overall mean GPA for all PSYC 210 classes for all time

Possible Results for Hypothesis Testing...

#1. H_0 (null hypothesis). There is **not** a significant difference between the means. The difference is just because of luck!

#2. H_1 (alternative hypothesis). There **is** a significant difference!

Possible Errors in Hypothesis Testing 🤖

	IN REALITY: H_0 is true (null hypothesis) <i>**In the real world there truly is <u>not</u> a significant difference**</i>	IN REALITY: H_0 is false (alternative hypothesis) <i>**In the real world there truly <u>is</u> a significant difference**</i>
YOUR DECISION: Reject H_0 (alternative hypothesis) <i>**You think there is a significant difference**</i>	Type I error (α , alpha) False Positive (you thought there was a real difference, but nope)	You're correct!
YOUR DECISION: Fail to reject H_0 (null hypothesis) <i>**You think there is <u>not</u> a significant difference**</i>	You're correct!	Type II error (β , beta) False negative (you thought it was nothing, but there is a real difference!)

BUT...how different is *significantly* different???? In other words, is my sample mean *far enough* away from the population mean that I should assume the sample came from a **different** population?

The next lesson answers this :)

Lesson 4: Hypothesis Testing with Z Tests

Z test...a test that determines if your sample mean is significantly different than your population mean 🙋

Example...Z tests can tell you if the mean GPA of psychology majors at UNC (your sample mean) is significantly higher than the average GPA of all UNC students (your population mean)

$$z = \frac{\bar{X} - \mu}{\frac{\sigma_x}{\sqrt{N}}}$$

Equation...

Z (Z score)

X with bar (sample mean)

μ (population mean)

σ (population standard deviation)

n (sample size)

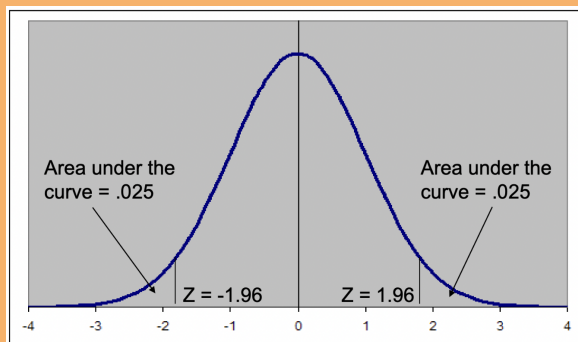
Alpha (α)...the cut-off line for determining if your results are significantly different. ✂️

Application...the typical alpha value is α = 0.05 which means that we only allow 5% of margin for error when determining if our results are significantly different

Remember...this is an *arbitrary number* chosen by psychologists. Your data doesn't affect your alpha value since psychologists set it beforehand.

*Note: Alpha is also the probability of a Type I error.

Critical Two-tailed Test...a test that allows a 2.5% margin of error *on each side* of the graph (adding up to 5% in total) in determining if your results are significantly different 2



Explanation of Z = ±1.96...this is the range of non-significant results. If your calculated Z statistic is in the “critical range.” i.e. greater than 1.96 on the right side (or less than -1.96 on the left side), then your results are significant. Yay!

Pros of two-tailed tests...

- (1) You can test errors on both sides
- (2) You have stricter criteria for determining if results are significant (you only allow 2.5% of error on a single side so your results have to be better to pass). Thus, there is less of a chance of a false positive

Cons of two-tailed tests...

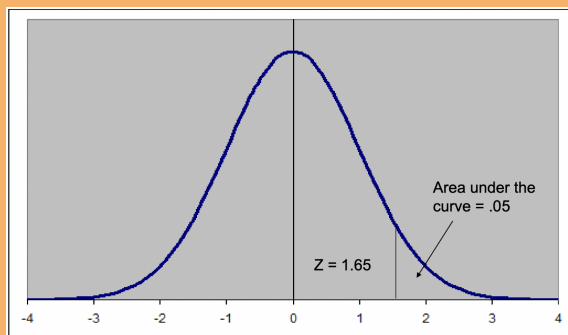
- (1) the strict guidelines make a false negative errors more likely since maybe there is an important difference but its doesn't pass the test

Note: we default to choosing two-tailed tests

P-value...the *likelihood* that there *really isn't* a significance difference given the results you have (aka the likelihood you're making a type I error) ✓

Explanation...a p-value is like doubt. If your p-value is under 0.01 that means there is 1% chance your results are not significant even though your results probably are significant (especially since any p-value under 0.05 is considered significant. You never know though!)

Critical One-tailed Test...a test that allows 5% margin of error *only on one side* of the graph in determining if your results are significantly different **1**



Pros of one-tailed tests...

- (1) Allows you to focus on significant results in a *specific* direction (in the positive direction or negative direction)
- (2) Makes false negatives less likely because there is more margin for error is 5% on one side rather than 2.5%. Maybe there *really is* a significant result but the data you got was just a bit off

Cons of one-tailed tests...

- (1) You can only measure significance on one-side
- (2) You are less strict with data you consider significant. With some data (like testing if a medical drug really works) it can be advantageous to be stricter.

Lesson 5: Hypothesis Testing with T tests

Power (1-β)...the probability that your data indicates a significant difference when *there really is a significance difference*, i.e. you're right! 🍀

How to increase your power...

- 1 use better test measures
- 2 use a homogenous population, i.e. collect data from people of similar political affiliation, gender, etc.
- 3 increase sample size

*note: power is the opposite of a type II error. A type II error occurs when your data indicates there is *not* a significant difference, even though in reality, there is a significant difference. Power says there is a significance difference when there truly is a significance difference.

Degrees of Freedom...sample size (basically). More specifically, it is the freedom each participant in your study has to vary. 🍀

Z tests are great...BUT for a Z test you need a population standard deviation...if we don't have that THEN... use a T test!

T test...a test that determines if your sample mean is significantly different than your population mean. BUT, this test doesn't require the population standard deviation to calculate. 🧐

Facts about the t-test...

- 1) Degrees of Freedom is n-1 which is sample size - 1.
- 2) Critical region changes based on your degrees of freedom, i.e. how different your sample mean needs to be to count as significant changes based on your sample size!
- 3) In the equation, use your sample standard deviation in place of the population standard deviation.

Pros...

- 1 Better for small samples because the t test is conservative as it takes sample size into account
- 2 you don't need a population standard deviation

Equation...
$$t = \frac{\bar{X} - \mu}{s_{\bar{X}}}$$

t = t statistic (your result)

X with bar (sample mean)

μ (population mean)

s_{X with bar} (the sample standard deviation divided by the square root of sample size)

***IMPORTANT!** You lose **1** degree of freedom with a t-test.

Which test????

If you have the population standard deviation...	If you <i>don't</i> have the population standard deviation...
Z TEST!	T TEST!

Independent Samples T test... a test that determines if groups you're comparing are significantly different 🤖

Example... run an experiment to determine if a group receiving a mood medication is significantly happier than the control group not receiving the medication

Important to know...

(1) this test measures differences between groups (control vs experimental) and within groups (differences within control group itself AND differences within experimental group itself)

(2) null hypothesis says the means of the samples are not significantly different, i.e. they come from the same population

$$\mu_1 = \mu_2$$

Equation...

$$t(df) = \frac{(\bar{X}_1 - \bar{X}_2)}{S_{\bar{X}_1 - \bar{X}_2}}$$

T (t test)

df (degrees of freedom)

X_1 with bar in numerator (first sample mean)

X_2 with bar in denominator (second sample mean)

The denominator is very complicated but, generally, is like randomness

*IMPORTANT! You lose **2** degrees of freedom with an independent samples T test.

*ALSO! The t statistic represents the ratio of *difference between groups / difference within groups*. In other words, the t statistic represents *systematic differences / random differences*.

Interpreting Results...

- 1) Calculate the t result
- 2) look at your giant table to critical values of t
- 3) Use the 0.5 column and pick the row associated with your degrees of freedom, i.e. if you have 20 degrees of freedom then use that row
- 4) locate the number at that row, column spot and that's your cut-off point
- 5) ask – is your t test score bigger or smaller than that cut-off point (If it's bigger than your results are significant. If it's smaller than your results are not significant).

Problems with T-tests! 😞

- (1) You must assume your original sample is from a *normal distribution*. But it could be abnormal!
- (2) Homosketacity. When you compare two samples, you assume the *differences within each of the individual samples* are the same. If you compare Dr. Harrison's morning class' grades to his afternoon class' grades we assume the differences within each class themselves are similar.
 - **Levene's Test for Equality of Variances** accounts for this! If your result for the test is not significant, that is good because it means the groups are adequately comparable. If you get a significant difference that indicates the differences within each sample are too extreme.
- (3) You must assume the samples are *independent* of each other. But what if you compare soccer vs basketball players and someone plays both sports!

Lesson 6: Regression & Correlation

Correlated Samples T test (aka Paired Samples T test; Dependent Samples T test)...a test that determines if two samples are significantly different. BUT, these samples are not independent of each other. The samples are related/overlap. 😞

Example...give students a pretest and posttest and see if the pre-scores are significantly different from the post-scores. The same students take the pre-test and post-test!

Types of samples...

- (1) Pretest/posttest
- (2) romantic couples
- (3) multiple interventions
- (4) related samples

*Note: You only lose 1 degree of freedom!

Linear relationship...the effect of one variable on another 🤖

Equation... $Y = bx + a$

a is y-intercept

b is slope

Regression equation...equation that shows how one independent variable (predictor variable) affects the dependent variable (outcome variable) 🌟

Example...does the number of hours you get affect your GPA? You could run a regression analysis where your independent variable is hours of sleep and your dependent variable is your GPA. In other words, you would ask: how does sleep predict GPA?

Two Equations...

$$\hat{Y}_i = bX_i + a$$

$$\hat{Y}_i = bX_i + a + e_i$$

$\hat{Y}$ is the predicted outcome

b is the slope

X_i is the individual's predictor score

a is the y-intercept

e is error/chance/randomness

Extra Terms...

- Beta is the standardized slope which is NOT in raw units. Beta tells you how much of an effect your dependent variables has.

- r is the correlation coefficient which indicates the strength of relationship between the variables

- Y (not $\hat{Y}$, just Y) is the actual outcome

- b is the unstandardized slope which is in raw units. B is the same as R for regression analysis with two variables.

Effect Size based on R-value 😊

Effect Size	R-value
Small	.10
Medium	.30
Large	.50

Line of best fit...line that minimizes deviations between $\hat{Y}$ (predicted outcome) and Y (actual outcome).

Aka, the line that is closest to the most points. ↗

Example...you run a regression analysis with a bunch of data scattered on a plot. The line of best fit is the “best guess” line that tries to get closest to all the dots.

Note #1...this is also called the “*least squares criterion*” which pretty much says: when you don't know how the variables relate to each other, the slope is the best guess

Note #2...the slope doesn't show you *how strong the variables relate*. Because the slope is in raw units (like shoe size, basketball points, etc.), it isn't standardized across psychology so you cannot compare it to other studies.

SO WHAT DO WE DO???

We transform the raw units into standardized Z scores

Standardized Z scores...when you want to find the relationship between two variables, you can run a regression analysis. But your results from that regression analysis are going to be in raw units (ex. If you study hours of sleep and GPA, your raw results are in hours and GPA units). SO – you want a standardized result, you have to convert your units to a common metric: standardized Z scores 

Notes...

- 1) A graph in standardized Z score units has the means set to 0, the standardized deviations set to 1

R²...the proportion of variance two variables share

Example...you have a midterm tomorrow, how do you get an A? You could eat healthy, sleep, study, etc. But how much do each of these variables explain the deviations in your midterm grade? R² answers that question.

Effect Size based on R²

Effect Size	R ²
Small Effect	0.01
Medium Effect	0.09
Large Effect	0.25

Correlation coefficient...the degree the *variable x* and *variable y* covary V.S. how much the *variable x* and *variable y* vary individually 😊

Example...you want to see how the number of hours you sleep affects your GPA. You could compare how the number of hours you sleep and GPA covary *together* (i.e. does more sleep = higher GPA?) versus how the number of hours you sleep and GPA covary individually (i.e. do you usually get the same number of hours of sleep every night or is there a lot of variability in the number of hours you sleep? Is your GPA usually consistent individually or does your GPA vary a lot semester-to-semester?).

This includes content up to midterm 1